



First Name: _____

Family Name: _____

Student Number: _____

Diploma of Engineering
University Entrance Program
Foundation Studies Program

DPEN022 – Enabling Mathematics B

Trigonometry Exam: Sample Solutions

This exam represents 20% of the total marks for the subject

Time allowed:	1 hour
Items permitted in exam:	UOW approved calculator
Aids supplied for exam:	Formula Sheet

INSTRUCTIONS TO STUDENTS

1. Write in blue or black pens only. Pencils are permitted for diagrams and graphs.
2. All questions are to be attempted.
3. All notation is as given in lectures.
4. Show all necessary WORKING and reasoning in the spaces provided.
5. Circle the letter (A, B, C, D, or E) corresponding to the correct answer to each question.
6. You must achieve 40% in the Trigonometry Exam. If this requirement is not met, your final mark for the subject will be a Technical Fail
7. Total number of questions: 15
 - Questions 1-6 are multiple choice
 - Each question is worth 1 mark
 - Marks are awarded for the answer only.
 - Questions 7-15 are long answer questions
 - The value of each question is indicated.
 - Part marks may be awarded.

No examination material may be removed from the examination venue.

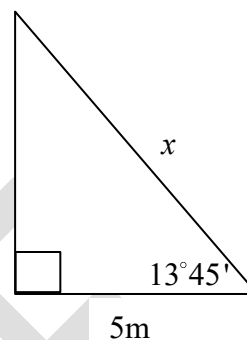
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Questions 1-6 are multiple choice. Show all necessary **WORKING** and reasoning in the spaces provided. Circle the letter (A, B, C, D, or E) corresponding to the correct answer to each question.

Question 1

In the right-angled triangle shown below, find the value of x , correct to 2 decimal places.

$$\begin{aligned}\cos 13^\circ 45' &= \frac{5}{x} \\ x \times \cos 13^\circ 45' &= 5 \\ x &= \frac{5}{\cos 13^\circ 45'} \\ &= 5.15\text{m}\end{aligned}$$



Option B

Question 2

What is the reference angle for $\theta = -135^\circ$.

$$\begin{aligned}\theta &= -135^\circ \\ &= -135^\circ + 360^\circ \\ &= 225^\circ \\ \theta_R &= 225^\circ - 180^\circ \\ &= 45^\circ\end{aligned}$$

Option E

Question 3

Determine the quadrant in which the angle θ must lie, given $\sec \theta < 0$ and $\cot \theta > 0$.

$$\sec \theta < 0 \Rightarrow \cos \theta < 0 \text{ is in Quadrant 2 and 3}$$

$$\cot \theta > 0 \Rightarrow \tan \theta > 0 \text{ is in Quadrant 1 and 3}$$

Quadrants overlap in Quadrant 3

Option C

Question 4

Convert the angle $\frac{11\pi}{12}$ into degrees.

$$\begin{aligned}\frac{11\pi}{12} &= \frac{11\cancel{\pi}}{12} \times \frac{180^\circ}{\cancel{\pi}} \\ &= 165^\circ\end{aligned}$$

Option A

Question 5

Write the equation for the resulting function $y = \cos x$ if the following transformations have been performed:

- vertically stretched so the amplitude is 3.
- horizontally stretched so that the period is 4π .
- shifted vertically down by 2 units.

$$\begin{aligned} \text{period} &= \frac{2\pi}{B} \\ 4\pi &= \frac{2\pi}{B} \\ B &= \frac{2\pi}{4\pi} \\ &= \frac{1}{2} \end{aligned}$$

$$\begin{aligned} A &= 3 \\ h &= 0 \\ k &= -2 \end{aligned}$$

$$\begin{aligned} y &= A \cos B(x - h) + k \\ y &= 3 \cos\left(\frac{x}{2}\right) - 2 \end{aligned}$$

Option B**Question 6**

Simplify the expression $(1 - \sin^2 x) \sec^2 x$.

$$\begin{aligned} (1 - \sin^2 x) \sec^2 x &= \cos^2 x \times \sec^2 x \\ &= \cos^2 x \times \frac{1}{\cos^2 x} \\ &= 1 \end{aligned}$$

Option C

Questions 7-15 are long answer questions. Show all necessary *WORKING* and reasoning in the spaces provided. The value of each question is indicated. Part marks may be awarded.

Question 7 (2 marks)

In a right-angled triangle, the hypotenuse is 15m and one side is 7m. Find the angle between the two given sides to the nearest minute.

$$\cos \theta = \frac{7}{15}$$

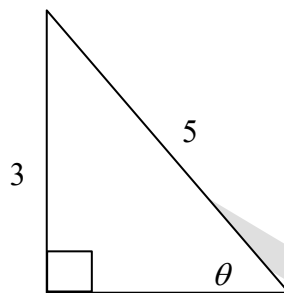
$$\begin{aligned}\theta &= \cos^{-1}\left(\frac{7}{15}\right) \\ &= 62^{\circ}10'54.7'' \\ &= 62^{\circ}11'\end{aligned}$$

Question 8 (3 marks)

If $\sin \theta = \frac{3}{5}$ and θ is acute.

- (i) Draw a diagram to represent this information.
- (ii) Find the exact value for $\sec \theta$.

(i)



(ii)

$$\begin{aligned}5^2 &= 3^2 + x^2 \\ x &= \sqrt{5^2 - 3^2} \\ &= 4 \\ \sec \theta &= \frac{\text{hyp}}{\text{adj}} \\ &= \frac{5}{4}\end{aligned}$$

Question 9 (2 marks)

Find the value of x if, $\sin 80^\circ = \cos(90 - 2x)^\circ$.

$$\sin 80^\circ = \cos(90 - 2x)^\circ$$

$$\cos(90 - 80)^\circ = \cos(90 - 2x)^\circ$$

$$\cos(10)^\circ = \cos(90 - 2x)^\circ$$

$$10 = 90 - 2x$$

$$2x = 80$$

$$x = 40^\circ$$

Question 10 (2 marks)

Find the exact value of $\tan \frac{2\pi}{3}$.

$$\tan \frac{2\pi}{3}$$

$\frac{2\pi}{3}$ is in the 2nd quadrant

$\tan$ is negative in the 2nd quadrant

$$\text{reference angle} = \pi - \frac{2\pi}{3} = \frac{\pi}{3}$$

$$\tan \frac{2\pi}{3} = -\tan \frac{\pi}{3} = -\sqrt{3}$$

Question 11 (3 marks)

Solve the equation $\cos^2 \theta - \frac{1}{4} = 0$ given $0^\circ \leq \theta \leq 360^\circ$. Give your answers in exact form, expressed in degrees.

$$\cos^2 \theta - \frac{1}{4} = 0$$

$$\cos^2 \theta = \frac{1}{4}$$

$$\cos \theta = \pm \sqrt{\frac{1}{4}}$$

$$\cos \theta = \pm \frac{1}{2}$$

cos is positive in Q1 and Q4

cos is negative in Q2 and Q3

$$\text{reference angle} = \cos^{-1}\left(\frac{1}{2}\right) = 60^\circ$$

$$\text{answer in Q1: } \theta = 60^\circ$$

$$\text{answer in Q2: } \theta = 180^\circ - 60^\circ = 120^\circ$$

$$\text{answer in Q3: } \theta = 180^\circ + 60^\circ = 240^\circ$$

$$\text{answer in Q4: } \theta = 360^\circ - 60^\circ = 300^\circ$$

$$\theta = 60^\circ, 120^\circ, 240^\circ, 300^\circ$$

Question 12 (3 marks)

Solve the equation $\cot \theta = \frac{1}{\sqrt{3}}$ given $0 \leq \theta \leq 2\pi$. Give your answers in exact form, expressed in radians.

$$\cot \theta = \frac{1}{\sqrt{3}}$$

$$\tan \theta = \sqrt{3}$$

tan is positive in Q1 and Q3

$$\text{reference angle} = \tan^{-1}(\sqrt{3}) = 60^\circ = \frac{\pi}{3}$$

$$\text{answer in Q1: } \theta = \frac{\pi}{3}$$

$$\text{answer in Q3: } \theta = \pi + \frac{\pi}{3} = \frac{4\pi}{3}$$

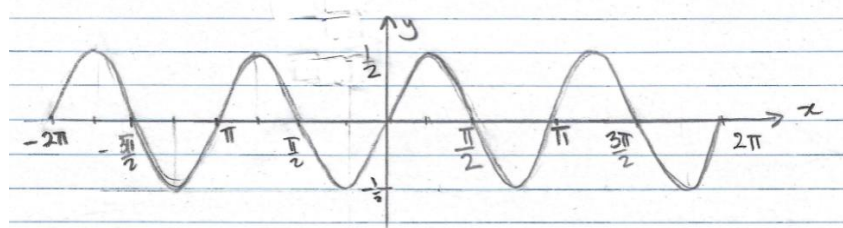
$$\theta = \frac{\pi}{3}, \frac{4\pi}{3}$$

Question 13 (3 marks)

Sketch the graph of $y = \frac{1}{2} \sin 2x$, for $-2\pi \leq x \leq 2\pi$ and state its amplitude and period.

$$\text{amplitude} = |A| = \left| \frac{1}{2} \right| = \frac{1}{2}$$

$$B = 2 \text{ period} = \frac{2\pi}{B} = \frac{2\pi}{2} = \pi$$

**Question 14 (3 marks)**

Prove $\frac{2 \cos^2 \theta}{1 - \sin \theta} = 2 + 2 \sin \theta$.

$$\begin{aligned} LHS &= \frac{2 \cos^2 \theta}{1 - \sin \theta} \\ &= \frac{2 \cos^2 \theta \times (1 + \sin \theta)}{(1 - \sin \theta) \times (1 + \sin \theta)} \\ &= \frac{2 \cos^2 \theta (1 + \sin \theta)}{1 - \sin^2 \theta} \\ &= \frac{2 \cos^2 \theta (1 + \sin \theta)}{\cos^2 \theta} \\ &= 2(1 + \sin \theta) \\ &= 2 + 2 \sin \theta \\ &= RHS \end{aligned}$$

Question 15 (3 marks)

Solve the equation $2 \sin^2 x - \cos x = 2$ for $0^\circ \leq x \leq 360^\circ$. Give your answers in exact form, expressed in degrees.

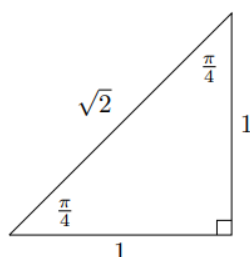
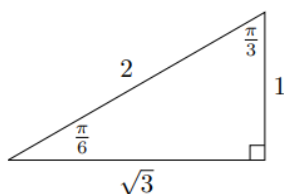
$$\begin{aligned} 2 \sin^2 x &= \cos x + 2 \\ 2(1 - \cos^2 x) &= \cos x + 2 \\ 2 - 2 \cos^2 x &= \cos x + 2 \\ 2 \cos^2 x + \cos x &= 0 \\ \cos x(2 \cos x + 1) &= 0 \\ \cos x = 0 \text{ or } 2 \cos x + 1 &= 0 \\ x = 90^\circ, 270^\circ, 120^\circ, 240^\circ \end{aligned}$$

END OF EXAM

Formula Sheet

Exact Trigonometric Ratios

$$\sin \theta = \frac{o}{h}, \quad \cos \theta = \frac{a}{h}, \quad \tan \theta = \frac{o}{a},$$



Acute Angles

$$\sin \theta = \cos(90^\circ - \theta)$$

$$\tan \theta = \cot(90^\circ - \theta)$$

$$\sec \theta = \operatorname{cosec}(90^\circ - \theta)$$

$$\cos \theta = \sin(90^\circ - \theta)$$

$$\cot \theta = \tan(90^\circ - \theta)$$

$$\operatorname{cosec} \theta = \sec(90^\circ - \theta)$$

Trigonometric Ratios

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\operatorname{cosec} \theta = \frac{1}{\sin \theta}, \quad \sin \theta \neq 0$$

$$\sec \theta = \frac{1}{\cos \theta}, \quad \cos \theta \neq 0$$

$$\cot \theta = \frac{1}{\tan \theta} = \frac{\cos \theta}{\sin \theta}, \quad \tan \theta \neq 0$$

Conversion Between Units

$$\text{Radians} \rightarrow \text{Degrees} \left(\times \frac{180^\circ}{\pi} \right)$$

$$\text{Degrees} \rightarrow \text{Radians} \left(\times \frac{\pi}{180^\circ} \right)$$

Graphs of Trigonometric Functions

$$y = A \cos B(x - h) + k$$

$$p = \frac{2\pi}{B}$$

Trigonometric Identities

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$$